

Ch-8

- # When two or more substances are mixed, where each substance retains its properties, it is known as a mixture. The individual substance by which a mixture is made is called a component.
- # The mixture, where components are visible with unaided eyes is called a ~~non-uniform~~ non-uniform mixture. But, a mixture, where components are not visible with unaided eyes is called a uniform mixture.
- # An alloy is a new kind of metal made mixing 2 or more metals in a fixed ratio. Eg - Stainless ~~stee~~ steel is made up of Iron, Nickel, Chromium & Carbon. Brass is made up of Copper & Tin.
- # Air is also a mixture as it consists of many gases. Eg - O_2 , CO_2 , N_2

Is CO_2 in air?

- ⇒ 1) Take a petri dish & mix CaO (calcium oxide) & H_2O (water) to make Ca(OH)_2 (calcium hydroxide)
- 2) Leave it for some time & return to it after a few hours.
- 3) It would turn milky because of formation CaCO_3 (calcium carbonate) which only occurs when lime water (Ca(OH)_2) comes in contact with CO_2 .
- 4) Hence, proved that CO_2 is in the air.

We learnt about SPM earlier but SPM is just a factor for ~~calc~~ calculating the AQI (Air Quality Index). AQI tells about the quality of air.

In Science, all the components of a mixture should be pure substances.

Pure & Adulterated

• A pure ~~sub~~ substance is a kind of matter that can't be separated into other kinds of matter by any physical method.

- A pure substance only consists of the same type of particles. Eg - Gold, O_2 , Gold.
- ~~Adulteration~~ Adulteration is an illegal process of adding substances which're cheaper or are of poor quality to a product. It might be done to increase quantity or reduce cost.

Separation of O_2 & H from water?

- 1) Fill a beaker $\frac{2}{3}$ rd with water & put a 9V battery in it.
- 2) Fill 2 small test tubes with beaker's water & without spilling place them on top of both the terminals of battery.
- 3) After some time you would see bubbles in both test tubes.
- 4) Now, take these test tubes & you would notice that they are filled with a gas.
- 5) The test tube on (+) terminal has O_2 , which can be tested by bringing a flame near it. The flame will burn brighter.
- 6) Tube on (-) terminal has Hydrogen, which can be tested using a flame, the gas burns with a 'pop' sound.

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~~Pure substance~~ Elements are pure ~~sub~~ substances which are made up of identical particles called atoms. Atoms can not be broken down further. Eg - Iron, Calcium

For atoms to stay in a stable form, they bond with ~~2~~ 2 or 3 such particles to make a molecule of that element. Eg - O_2

Elements can be metals, non-metal & metalloid (a substance having properties of both metals & non-metals)

-// Egs: Metals - Iron, Gold
Non-metal - Oxygen, Carbon
Metalloids - Silicon, Boron

There are 118 elements, most are solids, 11 are gaseous, 2 ~~are~~ (mercury & bromine) are liquid at room temp. 2 (gallium & caesium) are solids but liquid at room temp.

Room temp. is b/w 28° to $30^\circ C$ ($301-303K$)

Compound is formed when 2 or more elements are mixed in a certain ratio to form something completely new, whose properties are different from elements it was made from. Eg - NaCl (Sodium Chloride), Sugar

Constituent elements of a compound cannot be separated by any physical method.

When Iron & Sulphur are mixed & heated, they form iron sulphide, which is black in colour. When iron sulphide is mixed with H_2SO_4 , Iron ~~chloride~~ chloride is made along with a gas called Hydrogen sulphide. This gas has an egg-like odour. Iron sulphide is non-magnetic.

There is a substance called Graphene Aerogel. It is made from carbon. It's the lightest material on Earth. It's highly porous & is planned to be used as an environmental cleaner.

Mineral

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Mineral



Minerals are the substances found in rocks. They can be pure or a compound of 2 or more ~~an~~ element.

Mineral



They are found in compound form.
Eg- Calcite

Native Mineral



Minerals which are found in pure form.
Eg- Gold.